

DESCRIPTION OF FINAL ASSESSMENT

Course Code	CIC6314
Course Name	Artificial Intelligence
Term	2610
Academic Session	March/April 2026
Assessment Title	Group Project

A. Introduction/ Situation/ Background Information

This course introduces the fundamental principles of various learning techniques in Artificial Intelligence. It covers topics such as formulation of learning problems, search strategies, knowledge representation, logical reasoning, and their real-world applications. Students apply these concepts using tools such as **scikit-learn** through guided exercises and a group project. In this project, students work in teams to design and develop an intelligent system that solves a real-world decision-making problem by integrating core Artificial Intelligence techniques.

B. University Policy on Academic Misconduct

1. Academic Misconduct

Academic misconduct is a serious offense at Multimedia University Malaysia. It is defined as any of the following:

- i. Plagiarism – Submitting or presenting someone else’s work, words, ideas, data, or information as one’s own, whether intentionally or unintentionally. This includes incorporating published or unpublished material in manuscript, printed, or electronic form without proper acknowledgment of the source.
 - ii. Collusion – Two or more individuals collaborating on a piece of work (in part or in whole) that is intended to be completed individually, and presenting it as one’s own individual work.
 - iii. Cheating – Any act of dishonesty or fraud to gain an unfair advantage in an assessment. This includes using or attempting to use prohibited materials, assisting others in doing so, commissioning work from third parties, falsifying data, or violating examination rules.
2. All submitted assessments must be the student’s own original work, without the use of AI-

generated content. This includes direct copying, paraphrasing, or receiving assistance from AI tools. Any form of academic misconduct is a serious offense and will result in a zero mark for the affected assessment component or the entire assessment. In cases involving multiple parties (e.g., collusion), all involved parties will receive the same penalty.

C. Instruction to Students

In this group project, students are required to design and develop an intelligent AI system that can solve a real-world decision-making problem by integrating three core Artificial Intelligence components:

1. Search-Based Problem Solving
2. Knowledge Representation and Logical Reasoning
3. Machine Learning

The system must demonstrate how an AI agent can:

- i) Explore possible solutions using appropriate search strategies
- ii) Apply logical rules to support decision making
- iii) Learn patterns from data to improve prediction or classification

Application Domains (Choose ONE)

Students must select **one** of the following domains for their project:

- Medical Diagnosis Assistant
- Student Performance Prediction System
- Career Recommendation System
- Smart Product Recommendation System
- Travel or Route Planning Assistant

Submission type: A zip file that comprises Jupyter notebook, and poster.

Group Size: 4-5 members in a group.

Deadline for Project Submission: Week 13, 27/06/26, 7 PM

Presentation: Week 14

Only one team leader needs to perform the submission on behalf of the team. Please name the .zip using your team's name.

For the presentation, each group is given 10 minutes. No extra time will be given, so please plan your time wisely.

D. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Group Project (Code, Report)	20
2.	Group Project (Presentation)	10
	TOTAL	30

E. Task(s)

In this project, students are required to use an open dataset from a reputable source such as Kaggle or other public repositories. Students must carefully study the dataset description, understand its features, and evaluate its suitability for the chosen application domain. Students are required to form a group with classmates from the same lab group to work on this project. As this is a group project, proper project planning, task delegation, and effective communication among team members are essential to ensure the successful completion of high-quality work. Students are expected to collaborate closely in the formulation of the AI problem, the design of the data preparation and processing pipeline, the development of AI models, the experimental design, the analysis of results, and the documentation of the project findings.

Report

Please write your report and note using markdown cells in jupyter notebook, containing a precise description of the project. Most intermediate visualization and analysis should be provided in jupyter notebook. The report and presentation slides should include an Introduction, AI problem formulation, data preparation and processing pipeline, description of AI models applied, experiments, results, and discussion.

Code

Your code in jupyter notebook should be documented appropriately with explanations and justification of the analysis performed. The visualization should also be clearly described.

Presentation

Keep the poster presentation concise on what is actually being accomplished. Every team should present within the 10 min given. No extra time will be given, so please plan your time wisely. Each presentation is followed by 5 min Q&A.

APPENDIX 1

MARKING RUBRICS

Component Title	Group Project (Code, Report)					Percentage (%)	20	
Criteria	Score and Descriptors					Weight (%)	Marks	
	Excellent (9-10)	Good (7-8)	Average (5-6)	Need Improvement (3-4)	Poor (0-2)			
Search Algorithm Implementation	Correct and efficient implementation of BFS/DFS/A*. Clearly solves an optimal decision/path problem with heuristic (if A*).	Correct implementation; problem solved but lacks efficiency/clarity.	Basic implementation; limited demonstration of usefulness.	Incomplete or unclear implementation.	Not implemented or incorrect.	5		
Logic & Knowledge Representation	8–10 well-defined rules with correct inference mechanism. Clear reasoning demonstrated.	5–7 rules with working inference.	Few rules; reasoning partially works.	Rules present but inference unclear.	No proper rule-based reasoning.	5		
Machine Learning Model	Proper dataset, training/testing, evaluation. Correct model usage and interpretation.	Dataset and model used correctly with basic evaluation.	Model implemented but weak evaluation.	Minimal ML usage without clarity.	ML not implemented.	5		
Integration & System Design	Seamless integration: logic → search → ML → output. Clear workflow and design.	Components integrated with minor gaps.	Partial integration.	Components work separately.	No integration.	3		
Quality of Code & Report	Excellent coding practice, documentation, and very clear report with detailed explanation.	Good coding practice and clear report.	Acceptable code and basic report.	Poor documentation and unclear explanation.	Very poor code and report.	2		
TOTAL						20		

Component Title	Group Project (Presentation)					Percentage (%)	10
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (9-10)	Good (7-8)	Average (5-6)	Need Improvement (3-4)	Poor (0-2)		
Content Delivery and Q&A	Project content was delivered clearly with strong findings. Students can answer all questions by the lecturer and students.	Project content was delivered clearly with findings. Students can answer questions by the lecturer and students.	Project content was delivered clearly with findings. Students can answer most questions by the lecturer and students.	Project content was delivered clearly with inferior findings. Students can answer most questions by the lecturer and students.	Project content was not delivered clearly with inferior findings. Students cannot answer most questions by the lecturer and students.	5	
Teamwork & Peer Assessment	All team worked well together to achieve objectives. Each member contributed in a valuable way to the project since the group was formed.	Almost all team worked well together to achieve objectives. Each member contributed in a valuable way to the project.	The team worked well together most of the time, with only a few occurrences of communication breakdown or failure to collaborate when appropriate. Members were mostly respectful of each other	Few of the team worked together, with only a few occurrences of communication breakdown or failure to collaborate when appropriate. Few members were mostly respectful of each other	The team did not collaborate or communicate well since the group formed. Some members would work independently, without regard to objectives or priorities. A lack of respect and regard was frequently noted.	5	
TOTAL						10	

Note to students: Please print out and attach this appendix together with the submission of coursework